

TASK 10 – Compare Theory with Experimental Behaviour

The theoretical analysis predicted that **Linear Search** would have $O(n)$ growth, **Recursive Binary Search** would have $O(\log n)$ growth, and **Hash Search** would have $O(1)$ average-case growth. The experimental results obtained from the implementations generally support these theoretical predictions.

1. Linear Search

The theoretical complexity of Linear Search is:

$$\Theta(n)$$

Experimentally, the number of comparisons increased directly with the dataset size. For the **1,000,000-record dataset**, Linear Search required **1,000,000 comparisons**. This confirms the expected linear growth.

Therefore, the experimental behaviour agrees strongly with the theoretical analysis.

2. Binary Search

The theoretical complexity of Recursive Binary Search is:

$$\Theta(\log n)$$

For **1,000,000 records**, the experimental result required only **20 comparisons**, which is consistent with:

$$\log_2(1,000,000) \approx 20$$

Thus, the experimental result closely matches the theoretical prediction. Binary Search scales much better than Linear Search as n increases.

3. Hash Search

The theoretical average-case complexity of Hash Search is:

$$\Theta(1)$$

In the 1,000,000-record experiment, Hash Search required only **1 comparison**. This supports the expected constant average-case behaviour for a well-distributed hash table with no significant collision for the tested key.

However, actual Hash Search performance can vary depending on **collisions, hash-function quality, and table size**.

4. Which Algorithm Scales Better?

As the number of records increases:

- **Linear Search** becomes increasingly expensive because its comparisons grow with n .
- **Binary Search** remains efficient because its comparisons grow logarithmically.
- **Hash Search** provides the best average search performance when the hash table is properly designed.

Therefore, **Hash Search scales best on average**, followed closely by **Binary Search**, while **Linear Search scales the least effectively** for very large datasets.

5. Differences Between Theory and Experiment

There can be small differences between theoretical predictions and measured execution time. For example, the theoretical analysis focuses mainly on the number of operations and asymptotic growth, whereas actual execution time is affected by:

- Computer processor speed
- Memory access
- Compiler optimisation
- Operating-system activity
- Program overhead
- Hash-table construction and memory allocation

- Measurement precision of the timer

Therefore, execution time may not increase exactly according to the theoretical curve.

Conclusion

Overall, the experimental behaviour **agrees with the theoretical analysis**. Linear Search showed linear growth, Binary Search showed logarithmic growth, and Hash Search demonstrated approximately constant average search behaviour. The results confirm that **Binary Search and Hash Search are more suitable than Linear Search for large student-record datasets**, particularly when the number of records approaches or exceeds **1,000,000**.